

COMBINATORIAL CATEGORICAL EQUIVALENCES, TAKE 2

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1. INTRODUCTION

The purpose is to generalize our work in [15] concerning the Morita equivalence of partial maps and strong epimorphisms to include the Dold-Puppe-Kan Theorem [5, 6, 14] on additive simplicial objects and chain complexes.

2. THE SETTING

Let $\mathcal{P}$ be a category. Suppose $\mathcal{M}$ is a subcategory of $\mathcal{P}$ containing all the objects of $\mathcal{P}$ and containing all the isomorphisms. Suppose $\mathcal{S}$ is a class of morphisms such that every morphism f in $\mathcal{P}$ factors as $f = m \circ s$ uniquely up to isomorphism. We do not assume $\mathcal{S}$ is closed under composition. It does follow that $m = n \circ f$ with $m : U \rightarrow A$, and $n : V \rightarrow A$ in $\mathcal{M}$ implies $f \in \mathcal{M}$; we write $m \leq n$ when such an f exists.

Suppose there is a conservative faithful functor $(-)^* : \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$ which is the identity on objects and has $m^* \circ m = 1$ for all morphisms $m \in \mathcal{M}$. It follows that $m \leq n$ and $n \leq m$ imply m and n are isomorphic in the slice category $\mathcal{P}/A$. Write $\text{Sub}_{\mathcal{M}} A$ for the ordered set of isomorphism classes $[m]$ over X of morphisms $m : U \rightarrow A$ in $\mathcal{M}$. Write $\text{Subp}_{\mathcal{M}} A$ for the ordered set of isomorphism classes over X of noninvertible morphisms $m : U \rightarrow A$ in $\mathcal{M}$; the ‘p’ stands for ‘proper’.

Suppose $\mathcal{R}$ is a subclass of $\mathcal{S}$ with the following properties:

- (a) all the isomorphisms of $\mathcal{P}$ are in $\mathcal{R}$;
- (b) if u, r, v are composable with u and v invertible and $r \in \mathcal{R}$ then $v \circ r \circ u \in \mathcal{R}$;

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- (c) every morphism $s \in \mathcal{S}$ can be written uniquely as $s = r \circ m^*$ with $m \in \mathcal{M}$ and $r \in \mathcal{R}$;
- (d) if $m \in \mathcal{M}$ and $r \in \mathcal{R}$ is composable with m^* then $m^* \circ r \in \mathcal{S}$;
- (e) if $m^* \circ n \in \mathcal{R}$ with $m, n \in \mathcal{M}$ then $m^* \circ n$ is invertible and $m \circ m^* \circ n = n$;
- (f) if $r, r' \in \mathcal{R}$ are composable then $r' \circ r \in \mathcal{S}$;
- (g) if $s, t \in \mathcal{S}$, $m \in \mathcal{M}$ and $r \in \mathcal{R}$ with $t \circ s = m \circ r$ then $s \in \mathcal{R}$ and $t \in \mathcal{R}$.

We define a pointed category (that is, one whose homs have zero morphisms preserved by composition on either side) which we denote by $\mathcal{D}$. It has the same objects at $\mathcal{P}$ and the non-zero morphisms are those of $\mathcal{R}$. Composition is as in CP when the result is in $\mathcal{R}$ and zero otherwise.

Example 2.1. Take a category $\mathcal{A}$ with a factorization system $(\mathcal{E}, \mathcal{M})$. Assume that the pullback of any morphism with one in $\mathcal{M}$ exist. Assume every morphism in $\mathcal{M}$ is a monomorphism. Let $\mathcal{P}$ be the category $\text{Par}_{\mathcal{M}}\mathcal{A}$ of $\mathcal{M}$ -partial maps in $\mathcal{A}$. We have the functors $(-)_* : \mathcal{A} \rightarrow \mathcal{P}$ and $(-)^* : \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$ taking $f : A \rightarrow B$ to $[A \xleftarrow{1_A} A \xrightarrow{f} B]$ and $m : A \rightarrow B$ to $[B \xleftarrow{m} A \xrightarrow{1_A} A]$, respectively. Identify $\mathcal{M}$ in $\mathcal{A}$ with the morphisms m_* in $\mathcal{P}$ for $m \in \mathcal{M}$. Take $\mathcal{S}$ to consist of the partial maps of the form $e_* \circ m^*$ with $m \in \mathcal{M}$ and $e \in \mathcal{E}$. Take $\mathcal{R}$ to consist of the morphisms e_* in $\mathcal{P}$ with $e \in \mathcal{E}$.

Example 2.2. Take $\mathcal{P}$ to be the category $\Delta_{\perp, \top}$ of finite non-empty ordinals $n = \{0, 1, \dots, n-1\}$ with morphisms those functions which preserve first element, last element and order. We write $\sigma_k : m+1 \rightarrow m$ for the order-preserving surjection which takes the value k twice. We write $\partial_k : m \rightarrow m+1$ for the order-preserving injection which does not have k in its image. Take $\mathcal{M}$ to consist of all the injective functions in $\Delta_{\perp, \top}$ and take $\mathcal{S}$ to consist of all the surjective functions. Define $(-)^* : \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$ to be the functor which takes $\partial_k : m \rightarrow m+1$ to $\sigma_k : m+1 \rightarrow m$. (Note that $\sigma_k \dashv \partial_k$ in Cat .) Take $\mathcal{R}$ to consist of the isomorphisms and the surjections $\sigma_0 : m+1 \rightarrow m$.

3. THE TILDE CONSTRUCTION

Assume we are in the setting of Section 2.

Let $\mathcal{X}$ be an additive category with sufficient limits. There is a functor

$$\widetilde{(-)} : [\mathcal{P}, \mathcal{X}] \rightarrow [\mathcal{D}, \mathcal{X}]_{\text{pt}} \quad (3.1)$$

where the codomain category consists of the pointed functors. The definition on objects is:

$$\widetilde{T}A = \bigcap_{[U \xrightarrow{m} A] \in \text{Subp}_{\mathcal{M}} A} \ker T(m^* : A \rightarrow U) \quad (3.2)$$

and, for $r : A \rightarrow B$ in $\mathcal{R}$, the morphism $\widetilde{T}r : \widetilde{T}A \rightarrow \widetilde{T}B$ is the restriction of $Tr : TA \rightarrow TB$. Why does it restrict? Let $i_A : \widetilde{T}A \rightarrow TA$ be the inclusion. Take any $n : V \rightarrow B$ in $\text{Subp}_{\mathcal{M}} B$. Then $(Tn^*)(Tr)i_A = T(n^* \circ r)i_A = T(r' \circ m^*)i_A = (Tr')(Tm^*)i_A = 0$ using properties (c), (d) and (e) in Section 2. So there exists $\widetilde{T}r$ such that $i_B(\widetilde{T}r) = (Tr)i_A$, as claimed.

4. THE HAT CONSTRUCTION

We can also construct a functor

$$\widehat{(-)} : [\mathcal{D}, \mathcal{X}]_{\text{pt}} \longrightarrow [\mathcal{P}, \mathcal{X}] \quad (4.3)$$

whose value at the pointed functor $F : \mathcal{D} \longrightarrow \mathcal{X}$ is the functor $\widehat{F} : \mathcal{P} \longrightarrow \mathcal{X}$ defined as follows. On objects, put

$$\widehat{F}A = \sum_{[U \xrightarrow{m} A] \in \text{Sub}_{\mathcal{M}} A} FU .$$

The morphism $\widehat{F}f : \widehat{F}A \rightarrow \widehat{F}B$ has composite with the injection $\text{in}_m : FU \rightarrow \widehat{F}A$ ¹ equal to zero unless $s \in \mathcal{R}$ in which case it is the composite $\text{in}_n \circ Fs$ where we take the factorization $f \circ m = n \circ s$ with n in $\mathcal{M}$ and $s \in \mathcal{S}$. That is,

$$\widehat{F}f \circ \text{in}_m = \begin{cases} \text{in}_n \circ Fs & \text{for } s \in \mathcal{R} \\ 0 & \text{for } s \notin \mathcal{R} . \end{cases}$$

Why is $\widehat{F}$ a functor? Preservation of identities is clear since identities are in $\mathcal{R}$. Now take $f : A \rightarrow B$ and $g : B \rightarrow C$ in $\mathcal{P}$. Take $m : U \rightarrow A$ in $\mathcal{M}$. Factorize

$$f \circ m = n \circ s, \quad g \circ n = \ell \circ t, \quad t \circ s = n_1 \circ s_1$$

with $n, \ell, n_1 \in \mathcal{M}$ and $s, t, s_1 \in \mathcal{S}$. We first show that $s_1 \in \mathcal{R}$ if and only if $s, t, s \circ t \in \mathcal{R}$. By (f) of Section 2, if $s, t \in \mathcal{R}$ then $t \circ s \in \mathcal{S}$, and so, by uniqueness of $(\mathcal{S}, \mathcal{M})$ -factorisation, n_1 is invertible. If also $s \circ t \in \mathcal{R}$ then $s_1 \in \mathcal{R}$. Conversely, if $s_1 \in \mathcal{R}$, then, by (g) of Section 2, we have $s, t \in \mathcal{R}$; so n_1 is invertible and so $s \circ t \in \mathcal{R}$. Now we can calculate

$$\widehat{F}g \circ \widehat{F}f \circ \text{in}_m = \widehat{F}g \circ \text{in}_n \circ Fs = \text{in}_\ell \circ Ft \circ Fs$$

for $s \in \mathcal{R}$ and $t \in \mathcal{R}$, zero otherwise. By definition of $\mathcal{D}$ and functoriality of F , we have $Ft \circ Fs = F(t \circ s)$ for $t \circ s \in \mathcal{R}$, and $Ft \circ Fs = 0$ otherwise. So

$$\widehat{F}g \circ \widehat{F}f \circ \text{in}_m = \text{in}_\ell \circ F(t \circ s)$$

for $s \in \mathcal{R}$, $t \in \mathcal{R}$ and $t \circ s \in \mathcal{R}$, zero otherwise. From our first observation, this becomes

$$\widehat{F}g \circ \widehat{F}f \circ \text{in}_m = \text{in}_{\ell \circ n_1} \circ Fs_1$$

for $s_1 \in \mathcal{R}$, zero otherwise. That is,

$$\widehat{F}g \circ \widehat{F}f \circ \text{in}_m = \widehat{F}(g \circ f) \circ \text{in}_m$$

for all $[m] \in \text{Sub}_{\mathcal{M}} A$.

¹We abuse notation by writing in_m instead of $\text{in}_{[m]}$.

5. ADJOINTNESS

Theorem 5.1.

$$\widehat{(-)} \dashv \widetilde{(-)} : [\mathcal{P}, \mathcal{X}] \longrightarrow [\mathcal{D}, \mathcal{X}]_{\text{pt}}$$

Proof. We must prove that there is a natural isomorphism

$$[\mathcal{P}, \mathcal{X}](\widehat{F}, T) \cong [\mathcal{D}, \mathcal{X}]_{\text{pt}}[F, \widetilde{T}] .$$

Take a natural transformation $\theta : \widehat{F} \Rightarrow T : \mathcal{P} \rightarrow \mathcal{X}$. As in the definition of $\widehat{F}f$, with $f \circ m = n \circ s$ and $s \in \mathcal{R}$, we have commutativity in the following diagram.

$$\begin{array}{ccccc} FA & \xrightarrow{\text{in}_m} & \widehat{F}A & \xrightarrow{\theta_A} & TA \\ Fs \downarrow & & \downarrow \widehat{F}f & & \downarrow Tf \\ FB & \xrightarrow{\text{in}_n} & \widehat{F}B & \xrightarrow{\theta_B} & TB \end{array} \quad (5.4)$$

Consider a noninvertible $\ell : W \rightarrow A$ in $\mathcal{M}$. Then, by the properties of $\mathcal{R}$, $\ell^* \in \mathcal{S}$ and $\ell^* \notin \mathcal{R}$. So $T\ell^* \circ \theta_A \circ \text{in}_{1_A} = \theta_W \circ \widehat{F}\ell^* \circ \text{in}_{1_A} = \theta_W \circ 0 = 0$. This implies there exists a unique morphism $\phi_A : FA \rightarrow \widetilde{T}A$ such that the following square commutes.

$$\begin{array}{ccc} FA & \xrightarrow{\phi_A} & \widetilde{T}A \\ \text{in}_{1_A} \downarrow & & \downarrow i_A \\ \widehat{F}A & \xrightarrow{\theta_A} & TA \end{array} \quad (5.5)$$

Naturality of ϕ is proved as follows using (5.4) with $f = r \in \mathcal{R}$: $i_B \circ \phi_B \circ Fr = \theta_B \circ \text{in}_{1_B} \circ Fr = \theta_B \circ \widehat{F}r \circ \text{in}_{1_A} = Tr \circ \theta_A \circ \text{in}_{1_A} = Tr \circ i_A \circ \phi_A = i_B \circ \widetilde{T}r \circ \phi_A$.

For the inverse direction, take any natural transformation $\phi : F \Rightarrow \widetilde{T} : \mathcal{D} \rightarrow \mathcal{X}$. Define θ by commutativity of the following diagram.

$$\begin{array}{ccccc} FU & \xrightarrow{\text{in}_m} & \widehat{F}A & \xrightarrow{\theta_A} & TA \\ \phi_U \downarrow & & & & \uparrow Tm \\ \widetilde{T}U & \xrightarrow{i_U} & TU & & \end{array} \quad (5.6)$$

We need to prove the right-hand square of (5.4) commutes when precomposed with any in_m for any $m \in \mathcal{M}$. Put $f \circ m = n \circ s$ as usual. In the case where $s \in \mathcal{R}$, this follows from the following three commuting squares.

$$\begin{array}{ccccccc} FU & \xrightarrow{\phi_U} & \widetilde{T}U & \xrightarrow{i_A} & TU & \xrightarrow{Tm} & TA \\ Fs \downarrow & & \downarrow \widetilde{T}s & & \downarrow Ts & & \downarrow Tf \\ FV & \xrightarrow{\phi_V} & \widetilde{T}V & \xrightarrow{i_B} & TV & \xrightarrow{Tn} & TB \end{array} \quad (5.7)$$

In the case where $s \notin \mathcal{R}$, we can write $s = r \circ \ell^*$ for some $\ell \in \mathcal{M}$ noninvertible. Then $Tf \circ \theta_A \circ \text{in}_m = T(f \circ m) \circ i_U \circ \phi_U = T(n \circ r) \circ T\ell^* \circ i_U \circ \phi_U = T(n \circ r) \circ 0 \circ \phi_U = 0 = \theta_B \circ \widehat{F}f \circ \text{in}_m$, as required.

To show that the assignments are mutually inverse, take θ and define ϕ by (5.5). Let $\bar{\theta}$ be as θ is in (5.6). Then $\bar{\theta}_A \circ \text{in}_m = Tm \circ i_U \circ \phi_U = Tm \circ \theta_U \circ \text{in}_{1_U} = \theta_A \circ \widehat{F}m \circ \text{in}_{1_U} = \theta_A \circ \text{in}_m$. So $\bar{\theta} = \theta$.

On the other hand, take ϕ and define θ by (5.6). Let ϕ' be as ϕ is in (5.5). Then $i_A \circ \phi'_A = \theta_A \circ \text{in}_{1_A} = i_A \circ \phi_A$. So $\phi' = \phi$. $\square$

6. THE EQUIVALENCE

We assume now that $\text{Sub}_{\mathcal{M}} A$ is a finite set for some $A \in \mathcal{P}$. Every finite ordered set has a linear refinement. So we can choose representatives

$$m_0 : U_0 \rightarrow A, m_1 : U_1 \rightarrow A, \dots, m_p : U_p \rightarrow A$$

of all the $\mathcal{M}$ -subobjects of A such that $m_i > m_j$ implies $i < j$. We can take $U_0 = A$ and $m_0 = 1_A$.

Then the $\mathcal{M}$ -subobjects of U_i are all amongst the m_j for $j \geq i$. So we have a function $\rho : [p] \rightarrow \mathcal{P}[p]$ from $[p] = \{0, 1, \dots, p\}$ to the set of subsets of $[p]$ with the property that all the $\mathcal{M}$ -subobjects of U_i are represented by $m_{ij} : U_j \rightarrow U_i$ where $m_i \circ m_{ij} = m_j$ and $j \in \rho(i)$. We have $\rho(i) = \{\rho(i)_0 > \rho(i)_1 > \dots > \rho(i)_{\sigma(i)}\}$ where $\rho(i)_0 = i$.

Take a pointed functor $F : \mathcal{D} \rightarrow \mathcal{X}$. Take $0 < i \leq p$ and consider the morphism

$$\widehat{F}m_i^* : \widehat{F}A = FU_0 \oplus FU_1 \oplus \dots \oplus FU_p \rightarrow FU_{\rho(i)_0} \oplus FU_{\rho(i)_1} \oplus \dots \oplus FU_{\rho(i)_{\sigma(i)}} = \widehat{F}U_i.$$

Its composite with the j th inclusion is obtained by factoring $m_i^* \circ m_j = m_{i\rho(i)_k} \circ s$ with $s \in \mathcal{S}$. That composite is 0 unless $s \in \mathcal{R}$. However, using property (e) of Section 2, $m_{i\rho(i)_k}^* \circ m_i^* \circ m_j = s \in \mathcal{R}$ implies s invertible and $m_i \circ m_{i\rho(i)_k} \circ s = m_j$. Since we are dealing with one representative of each isomorphism class of $\mathcal{M}$ -subobjects, $s = 1_{U_j}$ and $\rho(i)_k = j$. So $\widehat{F}m_i^* \circ \text{in}_j = 0$ unless $j \leq i$ and there exists k with $j = \rho(i)_k$, in which case

$$\widehat{F}m_i^* \circ \text{in}_j = \text{in}_{\rho(i)_k}.$$

Write the inclusion $i_A : \widetilde{\widehat{F}}A \rightarrow \widehat{F}A$ as

$$i_A = (a_0, a_1, \dots, a_p)$$

where $a_i = \text{pr}_i \circ i_A : \widetilde{\widehat{F}}A \rightarrow FU_i$.

Lemma 6.1. $a_i = 0$ for $1 \leq i \leq p$.

Proof. We use induction. We have $a_p : \widetilde{\widehat{F}}A \rightarrow FU_p$ where U_p has no proper $\mathcal{M}$ -subobjects, giving $FU_p = \widehat{F}U_p$. So a_p is the restriction of $\widehat{F}m_p^* : \widehat{F}A \rightarrow \widehat{F}U_p$ to the subobject $\widetilde{\widehat{F}}A$ of its kernel. This gives the start $a_p = 0$ to our induction.

Assume $a_j = 0$ for $i < j \leq p$. Using the preamble to the Lemma, we see that $(a_i, a_{\rho(i)_1}, \dots, a_{\rho(i)_{\sigma(i)}})$ is the restriction of $\widehat{F}m_i^* : \widehat{F}A \rightarrow \widehat{F}U_i$ to the subobject $\widetilde{\widehat{F}}A$ of its kernel. By induction, $a_{\rho(i)_1} = \dots = a_{\rho(i)_{\sigma(i)}} = 0$. So $a_i = 0$, completing the proof. $\square$

Proposition 6.2. *The component at A of the unit $\eta_F : F \Rightarrow \widetilde{\widehat{F}}$ of the adjunction of Theorem 5.1 is invertible.*

Proof. The component $\eta_F A : FA \rightarrow \widehat{F}A$ of the unit is, using the above notation, defined by $i_A \circ \eta_F A = \text{in}_0$; it is clearly a monomorphism. Lemma 6.1 can be stated as saying $i_A = \text{in}_0 \circ a_0$. So

$$i_A \circ \eta_F A \circ a_0 = \text{in}_0 \circ a_0 = i_A = i_A \circ 1_{\widetilde{F}A} ,$$

yielding $\eta_F A \circ a_0 = 1_{\widetilde{F}A}$. Hence $\eta_F A$ has inverse $a_0 = \text{pr}_0$. $\square$

Now suppose $T : \mathcal{P} \rightarrow \mathcal{X}$ is any functor. We have the idempotents

$$e_i = Tm_i \circ Tm_i^*$$

on TA .

Lemma 6.3. *If the ordered set $\text{Sub}_{\mathcal{M}} A$ is finite with infima existing then the morphism*

$$\bigcap_{i=k}^p \ker e_i \oplus \bigoplus_{i=k}^p \widetilde{T}U_i \longrightarrow TA ,$$

induced by the inclusions $\ker e_i \rightarrow TA$ and the $Tm_i \circ i_{U_i} : \widetilde{T}U_i \rightarrow TA$, is invertible for $1 \leq k \leq p$.

Proof. For $k = p$ we have $\widetilde{T}U_p = TU_p$. The splitting $e_i = Tm_i \circ Tm_i^*$ yields the desired isomorphism $\ker e_p \oplus \widetilde{T}U_p \cong TA$.

For $k = p - 1$, the only possibility for the infimum of m_{p-1} and m_p is m_p . Then we have $m_{p-1,p} : U_p \rightarrow U_{p-1}$. So

$$\ker e_p = \ker Tm_p^* = \ker Tm_{p-1,p}^* \circ Tm_{p-1}^* \geq \ker e_{p-1} ,$$

yielding $\ker e_p \cap \ker e_{p-1} = \ker e_{p-1}$. We also have the split idempotent $e_{p-1,p} = Tm_{p-1,p} \circ Tm_{p-1,p}^*$ on TU_{p-1} yielding $TU_{p-1} \cong \ker Tm_{p-1,p}^* \oplus TU_p \cong \widetilde{T}U_{p-1} \oplus \widetilde{T}U_p$. So

$$TA \cong \ker e_{p-1} \oplus TU_{p-1} \cong \ker e_p \cap \ker e_{p-1} \oplus \widetilde{T}U_{p-1} \oplus \widetilde{T}U_p ,$$

as required.

And so on, inductively. However, a clearer proof will be given in the Appendix Section 7. $\square$

Theorem 6.4. *For any additive category $\mathcal{X}$ which has finite products and splittings for idempotents, the adjunction of Theorem 5.1 is an equivalence*

$$[\mathcal{P}, \mathcal{X}] \simeq [\mathcal{D}, \mathcal{X}]_{\text{pt}}$$

if the ordered set $\text{Sub}_{\mathcal{M}} A$ is a finite inf-semilattice for all objects A of $\mathcal{P}$.

Proof. After Proposition 6.2 it suffices to prove the counit invertible. However, case $k = 1$ of Lemma 6.3 gives the third isomorphism in the string

$$\widetilde{T}A \cong \widetilde{T}A + \sum_{i=1}^p \widetilde{T}U_i \cong \bigcap_{i=1}^p \ker e_i \oplus \bigoplus_{i=1}^p \widetilde{T}U_i \cong TA ,$$

and the composite is indeed the component of the counit at A . $\square$

7. APPENDIX: IDEMPOTENTS AND MEET-SEMI LATTICES

Let $\mathcal{X}$ be an additive category which has splittings for idempotents. Suppose e is an idempotent on an object A of $\mathcal{X}$. A splitting for e will be denoted as follows:

$$\begin{array}{ccc} A & \xrightarrow{e} & A \\ r_e \downarrow & \nearrow i_e & \downarrow r_e \\ eA & \xrightarrow{1_{eA}} & eA \end{array} \quad (7.8)$$

We also have the idempotent $\bar{e} = 1_A - e$ on A yielding a direct sum situation:

$$\bar{e}A \xrightleftharpoons[r_{\bar{e}}]{i_{\bar{e}}} A \xrightleftharpoons[i_e]{r_e} eA. \quad (7.9)$$

Consequently, we have an isomorphism

$$A \cong \bar{e}A \oplus eA.$$

Now suppose we have another idempotent f on A which commutes with e . It follows that $fe = f \circ e = e \circ f$ is also an idempotent on A . Moreover, f induces an idempotent on eA whose splitting is also a splitting feA for fe . This gives the following 3×3 -diagram of split exact sequences:

$$\begin{array}{ccccc} \bar{f}\bar{e}A & \longrightarrow & \bar{e}A & \longrightarrow & f\bar{e}A \\ \downarrow & & \downarrow & & \downarrow \\ \bar{f}A & \longrightarrow & A & \longrightarrow & fA \\ \downarrow & & \downarrow & & \downarrow \\ \bar{f}eA & \longrightarrow & eA & \longrightarrow & feA \end{array}$$

Consequently, we have an isomorphism

$$A \cong \bar{f}\bar{e}A \oplus f\bar{e}A \oplus \bar{f}eA \oplus feA.$$

Now suppose we have a family $(e_i)_{i \in I}$ of idempotents e_i on A which commute with each other. For each subset S of I , let e_S be the composite of the idempotents x_i , for all $i \in I$, where

$$x_i = \begin{cases} e_i & \text{for } i \in S \\ \bar{e}_i & \text{for } i \notin S \end{cases}.$$

A simple induction proves:

Lemma 7.1 (Idempotents Cube Lemma).

$$A \cong \bigoplus_{S \subseteq I} e_S A$$

We can order the set I by defining $i \leq j$ when $e_i e_j = e_i$. In this case, notice that $e_i \bar{e}_j = e_i - e_i e_j = 0$; so $e_i \bar{e}_j A = 0$. This shows that some terms in the direct sum in Lemma 7.1 could be zero.

Now suppose we have a finite lattice L ; that is, L is an ordered set which, when regarded as a category, has finite limits. (A finite ordered set with finite products has

finite coproducts too and so is a lattice.) As usual, we write $x \wedge y$ for the meet (= infimum = product) of x and y . The largest element of L is denoted by t . We can construct the usual category $\text{Spn}L$ whose objects are the elements $x \in L$ and whose morphisms $u : x \rightarrow y$ are elements $u \in L$ such that $u \leq x \wedge y$. Composition in $\text{Spn}L$ is meet.

Each endomorphism $u : x \rightarrow x$ in $\text{Spn}L$ is an idempotent and all these idempotents on x commute since meet is commutative.

Suppose we are given a functor $T : \text{Spn}L \rightarrow \mathcal{X}$. Put $A = Tt$. Each $x \in L$ gives an idempotent $e_x = T(x : t \rightarrow t)$ on A . This gives us a family $(e_x)_{x \in L}$ of commuting idempotents on A to which Lemma 7.1 applies. Notice that $x \leq u$ implies $e_x e_u = e_x$, $\bar{e}_x \bar{e}_u = \bar{e}_u$ and $e_x \bar{e}_u = 0$.

Lemma 7.2. *If $e_S \neq 0$ then S is a filter and $e_S = e_z d_z$, where $z = \bigwedge_{x \in S} x$ and d_z is the composite of all the idempotents $\bar{e}_u$ with $u < z$.*

Proof. Suppose $x \leq u$ with $x \in S$. Assume $u \notin S$. Then e_S factors through $e_x \bar{e}_u = 0$, so $e_S = 0$, a contradiction. So $u \in S$ proving that S is a right ideal. Now suppose $x, y \in S$ and $x \wedge y \notin S$. Then e_S factors through $e_x e_y \bar{e}_{x \wedge y} = e_{x \wedge y} \bar{e}_{x \wedge y} = 0$, a contradiction. So S is a filter.

Every filter on a finite set is principal: $S = \{x \in L : z \leq x\}$ for z as in the statement of the Lemma. Then e_z is the composite of the e_x with $x \in S$. Moreover, if $u \notin S$ and $u \not\leq z$ then we have $u \wedge z < z$ and $\bar{e}_u \bar{e}_{u \wedge z} = \bar{e}_{u \wedge z}$. So d_z is the composite of the $\bar{e}_u$ for $u \notin S$. This yields $e_S = e_z d_z$. $\square$

As a consequence, Lemma 7.1 gives

$$A \cong \bigoplus_{x \in L} d_x e_x A. \quad (7.10)$$

For $u < x$, notice that $K_{u,x} = \bar{e}_u e_x$ is the kernel of $Tu : Tx \rightarrow Tx$. This gives

$$\tilde{T}x = \bigcap_{u < x} K_{u,x} = d_x e_x A, \quad (7.11)$$

and so

$$Tt \cong \bigoplus_{x \in L} \tilde{T}x. \quad (7.12)$$

Applying this to $L = \text{Sub}_{\mathcal{M}} A$, we obtain Lemma 6.3 in the case of Example 2.1. For the general case we need further work.

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